

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
KIRUMAMPAKKAM, PUDUCHERRY-607 403.

B.Tech., Ist Year Sem-I - PHYSICS – Model Exam-February 2023

Section: A, B&C
Date: 11.02.2023

Max. Marks: 75
Time: 3 hrs

SECTION A (10x2=20marks)

Answer ALL the questions

All Questions carry equal marks

1. What are ultrasonic waves and mention two applications?
2. What is Acoustic grating?
3. What is the need to do anti reflection coating?
4. How will you construct quarter wave plate to produce polarized light?
5. Define Total internal reflection.
6. A fiber has core refractive index of 1.42 and cladding of 1.38. Determine the critical angle and numerical aperture.
7. State de Broglie hypotheses.
8. Define Quantum Mechanical Tunneling.
9. Define multiplication factor.
10. Distinguish between fission and fusion.

SECTION B(5X11=55marks)

Answer FIVE questions in “Either or” Pattern

All questions carry equal marks

UNIT I

11. Write the principle, construction and production of ultrasonic waves by Piezo electric method with neat circuit diagram. **(11)**
- OR**
12. Derive the formula for reverberation time using Sabine’s concept. **(11)**

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SECTION B(5X11=55marks)

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UNIT I

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- OR**
12. Derive the formula for reverberation time using Sabine’s concept. **(11)**

UNIT II

13. Explain the construction, working and types of fringes in Michelson interferometer and hence explain how it is used to find the wavelength of monochromatic source (11)

OR

14. a. Define diffraction of light.. (4)
b. Derive an expression for resolving power of grating. (7)

UNIT III

15. Explain the Principle, construction and working of CO₂ laser with energy level diagram.

OR

- 16 a. Find the relation between acceptance angle and numerical aperture of the fiber. (5)
b. Explain the fiber optic communication system. (6)

UNIT IV

17. Derive the Schrödinger time independent and dependent equations. (11)

OR

18. With neat diagram explain the V-I characteristics of Tunnel Diode. (11)

UNIT V

19. Explain the P- P and C - N cycle, D-D and D-T thermonuclear reactions. (11)

OR

20. Explain the Principle, construction, working, merits and demerits of FBTR with neat diagram (11)

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All the Best***